Status Report: Addressing Class Imbalance in Skin Cancer Datasets

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Overview

Current Dataset and Data Leakage

Preprocessing Oversights

Model Evaluation and Limitations

Conclusion and Future Work

Dataset and Data Leakage

Feature Extraction

ORIGINAL EXTRACTED FEATURES

- **Average RGB color values** (avg_r, avg_g, avg_b)
- **Color histograms** — pixel intensity distribution across R, G, B channels (96 features)
- **Haralick texture features** — contrast, homogeneity, entropy from GLCM (13 features)
- **Edge detection** — Canny, Sobel, Laplacian, Gaussian filters (12 features)
- **Shape features** — enclosing circle radius/area, bounding rectangle width/height/ratio (6 features)

ENGINEERED FEATURES

- **Channel ratios** — R/G, R/B, G/B color dominance ratios
- **Color contrast** — R-G, R-B, G-B channel differences
- **Color moments** — brightness, color range, color std
- **ABCD shape features** — asymmetry ratio, border irregularity, diameter estimate

Data Integrity

Validation Strategy

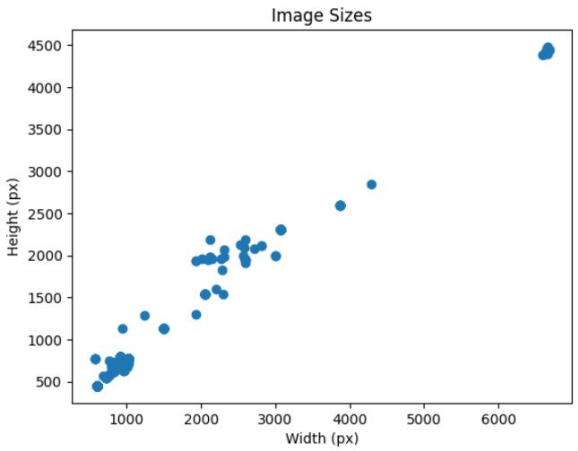
- Move images from training pool to test set to ensure equal classes
- Split remaining training data 80/20 into train and validation sets using stratified sampling
- Test set kept completely separate as final holdout for unbiased evaluation

Preventing Data Leakage

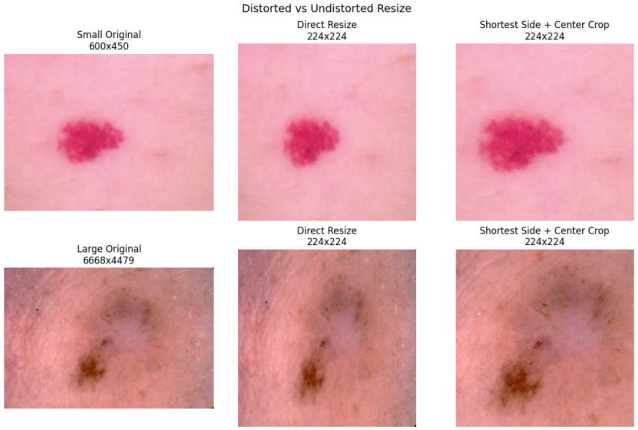
- Keep all directories separate to ensure images don't accidentally cross over
- Split training data into train/val before preprocessing
- Apply preprocessing to each split independently after splitting

Size Problem

Image sizes vary heavily, leading to larger images compression when directly resized to 224,224. Lesions and edges of images are distorted, leading to CNN model becoming confused



Resizing the shortest side of the image to 256 and taking a centercrop of 224x224 eliminates edges while maintaining focus on the lesion



Stratified Split

Regular splitting led to inconsistent class distributions in training and validation. Leads to high variance in metrics given limited data points

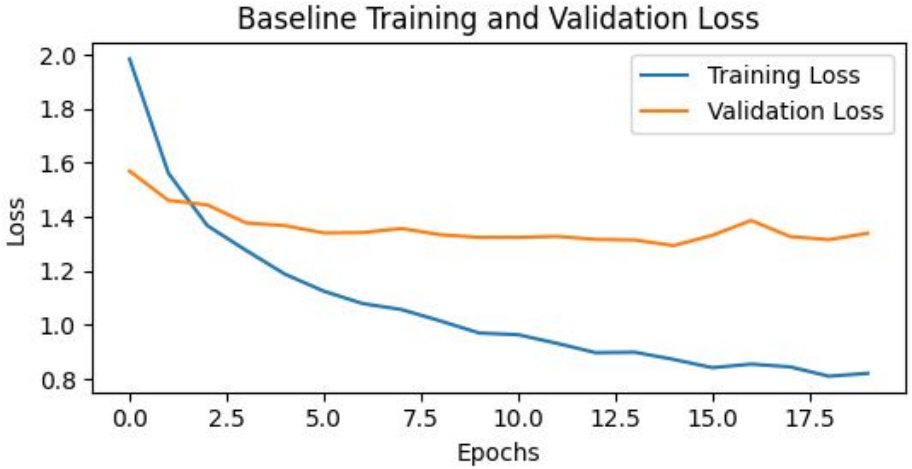
Class	Original Train	Naive Train	Naive Val	Stratified Train	Sratified Val
Vascular Lesion	5.69%	5.54%	6.32%	5.66%	5.82%
Pigmented Benign Keratosis	20.88%	20.17%	**23.70%**	20.89%	20.81%
Squamous Cell Carcinoma	8.18%	8.70%	**6.09%**	8.15%	8.28%
Melanoma	19.79%	19.21%	**22.12%**	19.82%	19.69%
Dermatofibroma	4.29%	4.58%	3.16%	4.3%	4.25%
Nevus	16.13%	16.27%	15.58%	16.14%	16.11%
Seborrheic Keratosis	2.89%	3.28%	**1.35%**	2.89%	2.91%
Actinic Keratosis	5.15%	5.03%	5.64%	5.15%	5.15%
Basal Cell Carcinoma	16.99%	17.23%	16.03%	16.99%	17%

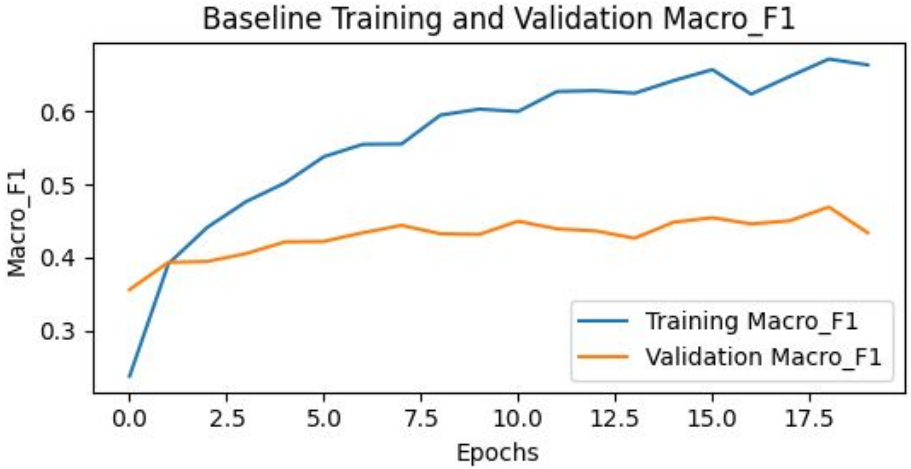
MobileNetV2

F1-Score and Loss Curves

Moderate gap between curves and relatively stagnant validation loss and macro F1-score indicate overfitting

- Even at 20% split, some classes are as small as 13 images. Small misclassifications can lead to high variance in validation metrics





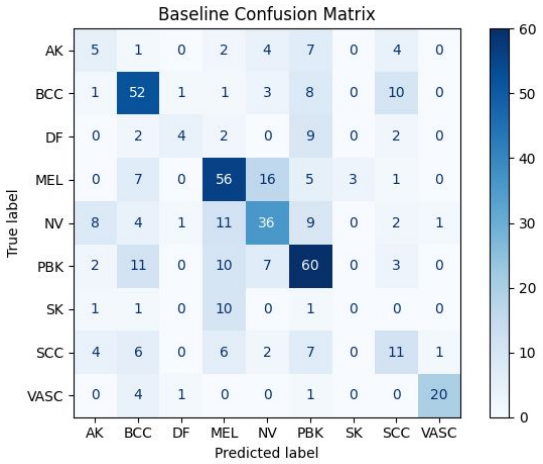
Report and Confusion Matrix

Seborrheic Keratosis failed to be classified completely

- Other minority classes such as dermatofibroma and actinic keratosis also performed poorly

Majority classes had the highest F1-scores generally

- Notable exception is vascular lesion (smaller class but an 83% F1-score)

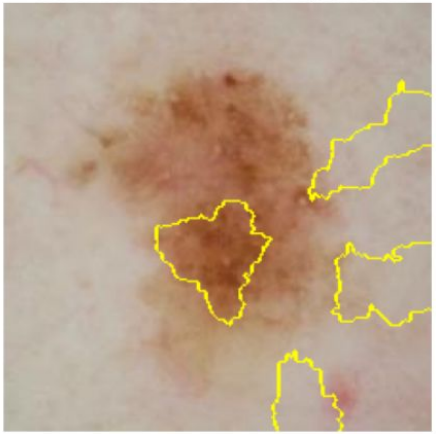


Baseline Classification Report

	precision	recall	f1-score	support
AK	0.24	0.22	0.23	23
BCC	0.59	0.68	0.63	76
DF	0.57	0.21	0.31	19
MEL	0.57	0.64	0.60	88
NV	0.53	0.50	0.51	72
PBK	0.56	0.65	0.60	93
SK	0.00	0.00	0.00	13
SCC	0.33	0.30	0.31	37
VASC	0.91	0.77	0.83	26
accuracy			0.55	447
macro avg	0.48	0.44	0.45	447
weighted avg	0.53	0.55	0.53	447

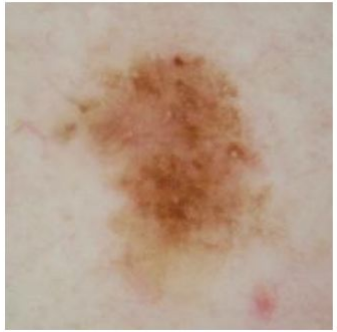
LIME and Grad-CAM

Seborrheic Keratosis LIME Explanation

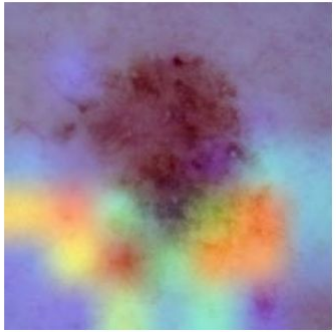


Vascular Lesion LIME Explanation

Original
True: SK

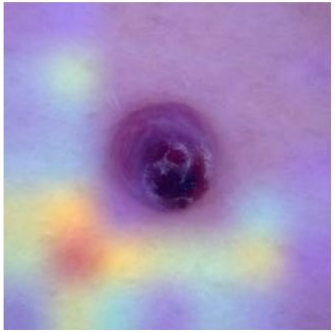


Grad-CAM
Pred: melanoma (67.44%)



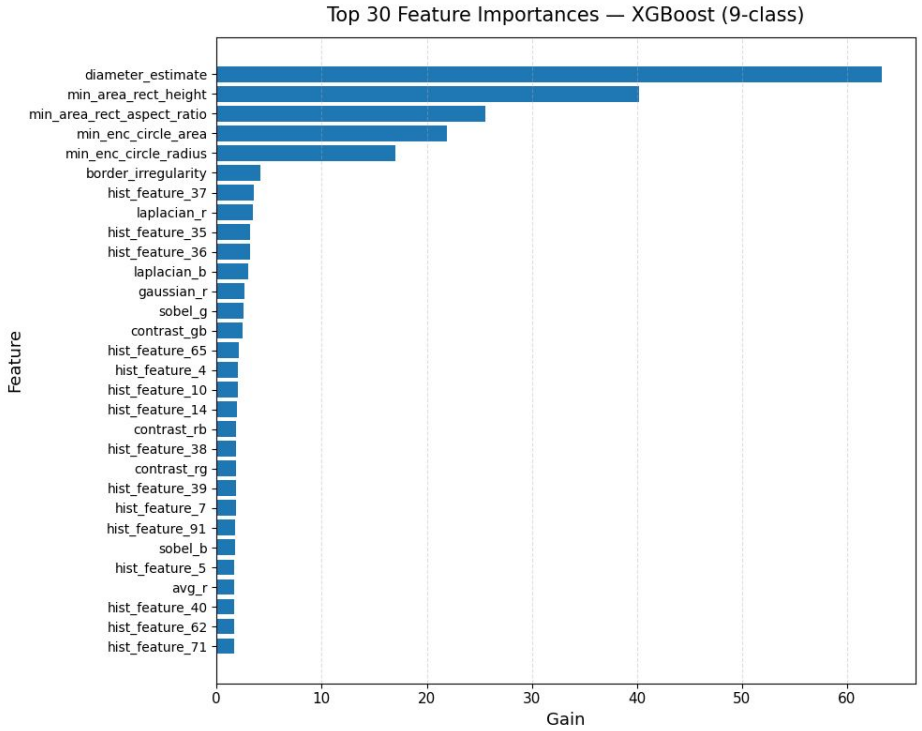
Original
True: Vascular Lesion

Grad-CAM
Pred: vascular lesion (80.40%)



XGBoost

Preprocessing + Feature engineering



- Stratified 80/20 train/test split
- No missing values — 2,357 samples, 130 original features
- Added 12 engineered features — channel ratios, color contrast, color moments, ABCD shape features (142 total)

Class Imbalance Handling

```
Before SMOTE:
label
0    382
1    363
2    313
3    298
4    158
5    114
6    104
7     89
8     64
Name: count, dtype: int64

After SMOTE:
label
0    382
1    382
2    382
3    382
4    382
5    382
6    382
7    382
8    382
Name: count, dtype: int64

X_train shape after SMOTE: (3438, 142)
```

	Approach	Test Accuracy	F1 macro	Seborrheic F1	Vascular F1	Melanoma F1
0	SMOTE	0.6038	0.50	0.00	0.76	0.66
1	SMOTETomek	0.5975	0.50	0.06	0.74	0.68
2	Class Weights	0.6038	0.51	0.00	0.79	0.67

- Dataset imbalanced — pigmented benign keratosis (478) vs seborrheic keratosis (80)
- Tested 3 approaches — SMOTE, SMOTETomek, class weights
- Class weights best overall F1 macro (0.51)
- SMOTETomek only approach to detect seborrheic keratosis at all (0.06)

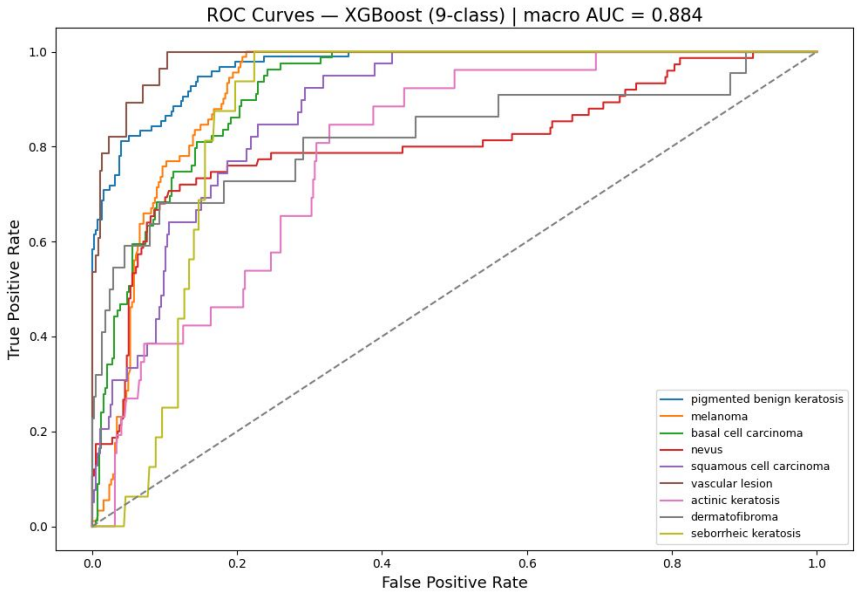
Hyperparameter Tuning

```
Fitting 5 folds for each of 40 candidates, totalling 200 fits

BEST PARAMETERS:
{
  "xgb__subsample": 0.8,
  "xgb__reg_lambda": 1.0,
  "xgb__reg_alpha": 0.0,
  "xgb__n_estimators": 300,
  "xgb__max_depth": 5,
  "xgb__learning_rate": 0.1,
  "xgb__colsample_bytree": 1.0
}

Best CV F1 macro: 0.7969
```

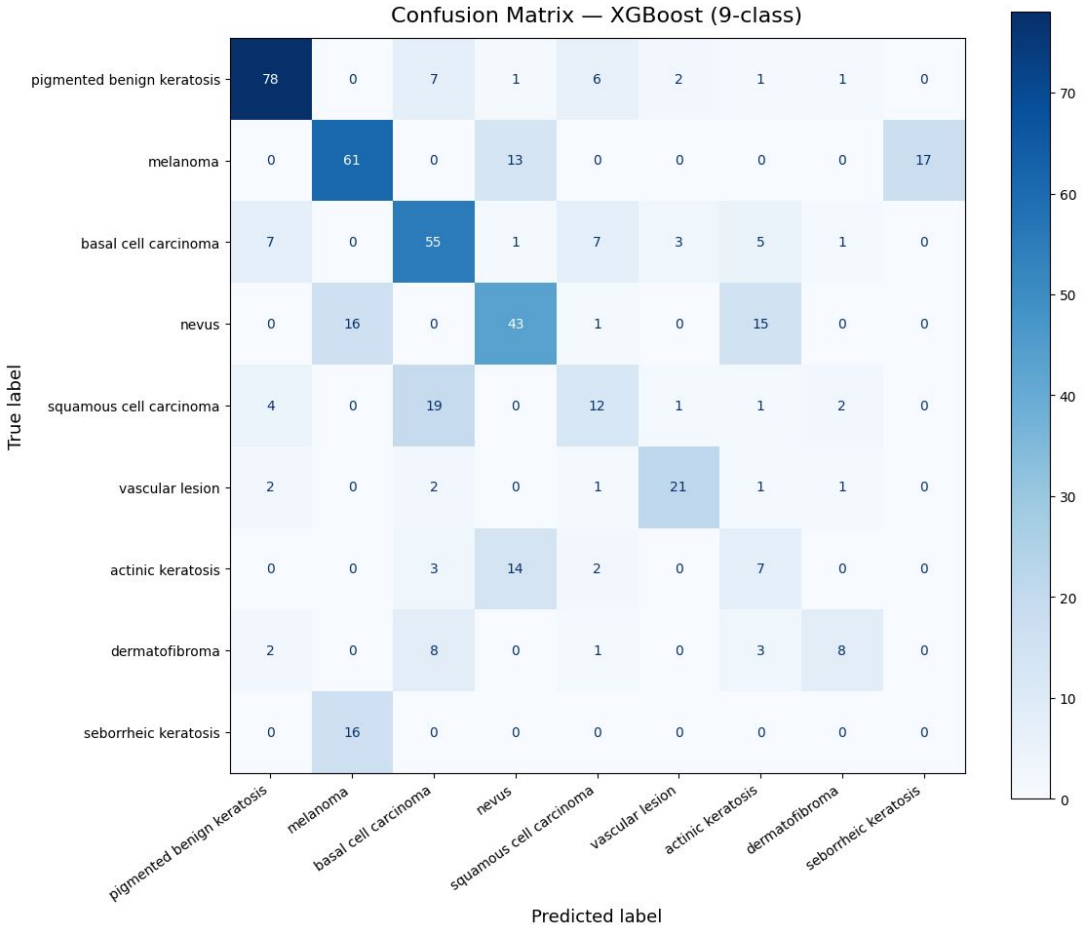
- RandomizedSearchCV with 5-fold StratifiedKFold
- 40 iterations, optimizing F1 macro
- Best CV F1 macro: 0.7969
- Best params: 400 estimators, max depth 5, learning rate 0.1

Model Evaluation

```
Test Accuracy: 0.6038

Classification Report (9-class XGBoost):
                          precision    recall  f1-score   support

pigmented benign keratosis       0.84      0.81      0.83        96
                  melanoma       0.66      0.67      0.66        91
      basal cell carcinoma       0.59      0.70      0.64        79
                     nevus       0.60      0.57      0.59        75
   squamous cell carcinoma       0.40      0.31      0.35        39
          vascular lesion        0.78      0.75      0.76        28
        actinic keratosis        0.21      0.27      0.24        26
           dermatofibroma        0.62      0.36      0.46        22
      seborrheic keratosis       0.00      0.00      0.00        16

                 accuracy                           0.60       472
                macro avg       0.52      0.49      0.50       472
             weighted avg       0.61      0.60      0.60       472
```

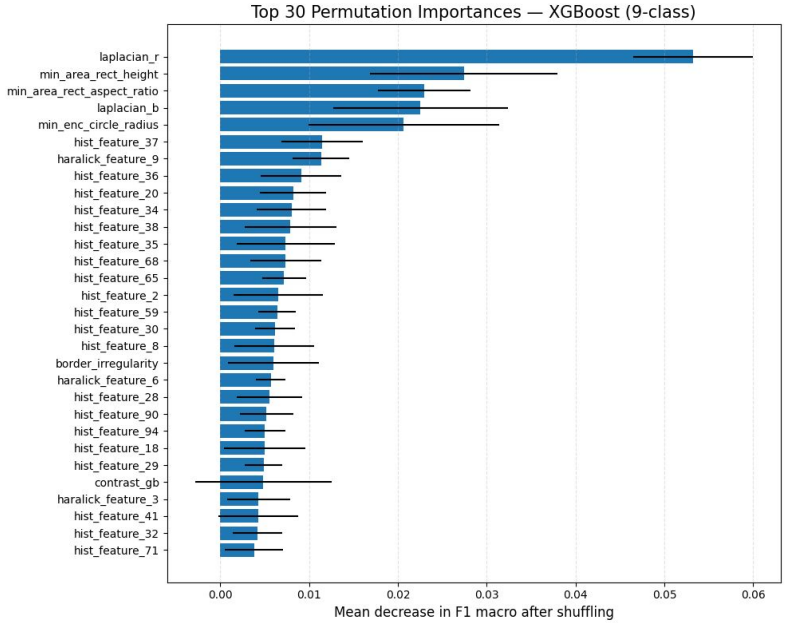


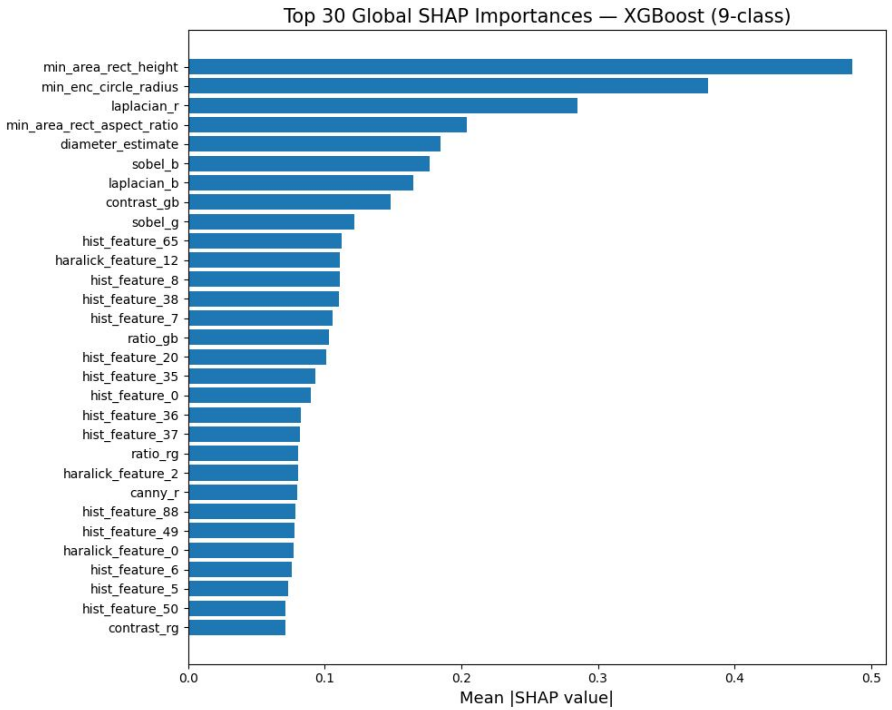
- Test accuracy: 60.4%
- ROC-AUC macro: 0.884
- F1 macro: 0.51
- Strongest: vascular lesion (0.76), pigmented benign keratosis (0.83)
- Weakest: seborrheic keratosis (0.00), actinic keratosis (0.24)



- Squamous cell carcinoma most confused with basal cell carcinoma (19 times)
- Seborrheic keratosis misclassified as melanoma all 16 times
- Visually similar classes consistently confused with each other
- Strong diagonal for pigmented benign keratosis and melanoma

Explainability

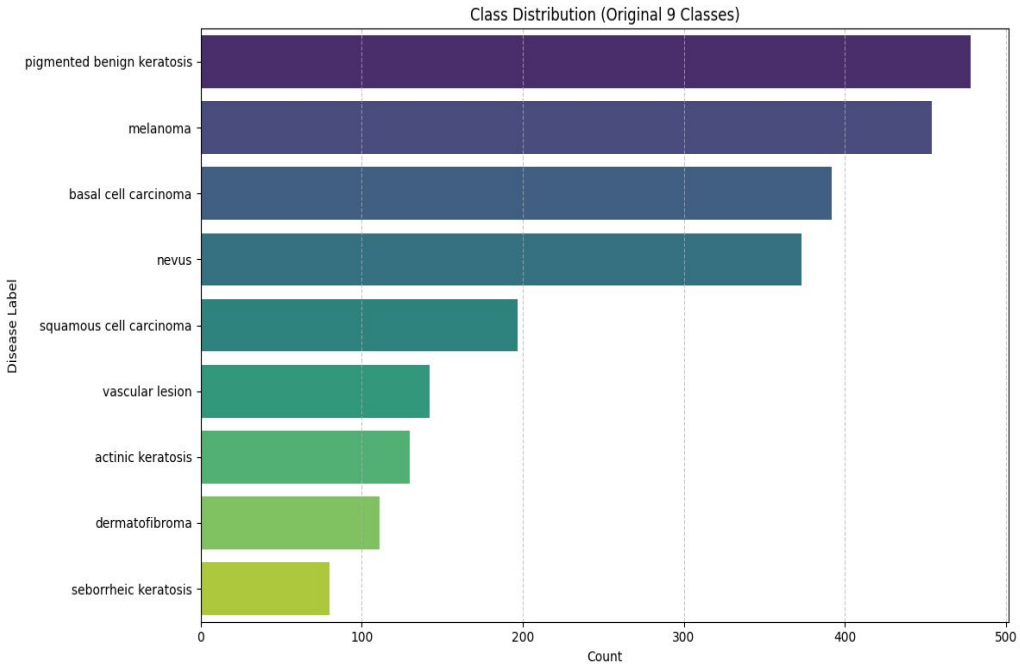




- Engineered features dominated top rankings in both SHAP and permutation importance
- Diameter estimate and border irregularity ranked top 5
- Color contrast and ratio features in top 30
- Model relies on shape and color over texture

Random Forest

Random Forest



Baseline Model

- Accuracy : 36%, Micro F1 : 31%
- Misclassified Actinic Keratosis & Seborrheic Keratosis Completely

Preprocessing

- No Missing Values
- Label Encoding
- Train / Val / Test - 80/10/10 or 70/15/15
- Smote & Borderline Smote
- Hyperparameter Tuning

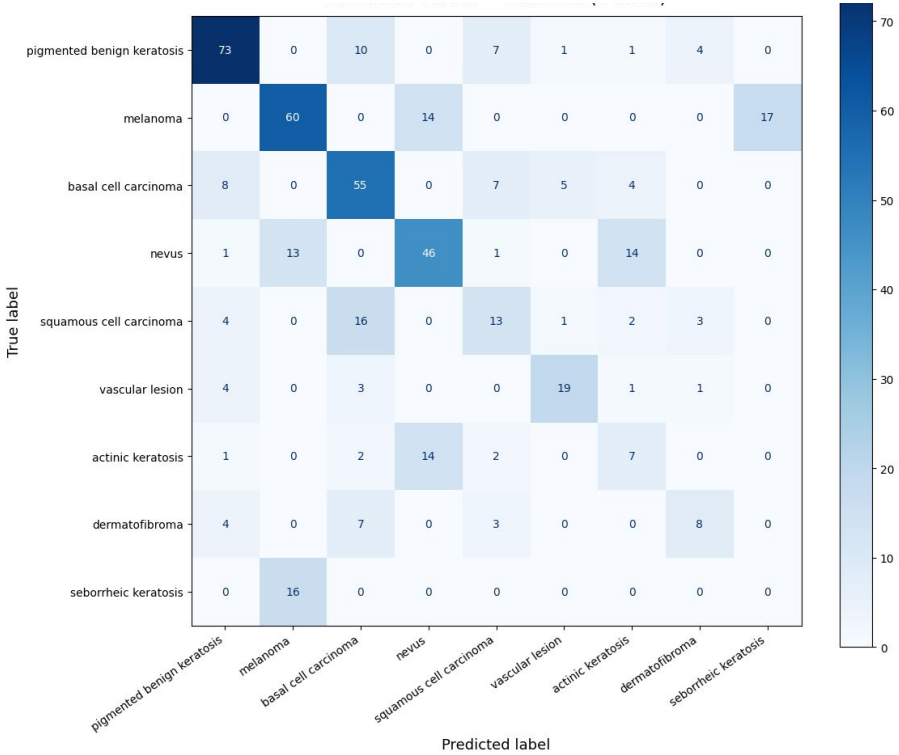
Random Forest - Tuning

```
BEST PARAMETERS:
{
  "rf__n_estimators": 300,
  "rf__min_samples_split": 2,
  "rf__min_samples_leaf": 1,
  "rf__max_features": "sqrt",
  "rf__max_depth": 30
}

Best CV F1 macro: 0.7721
```

Hyperparameters

- Used **RandomizedSearchCV with 5-fold Stratified Cross-Validation**
- Optimized using **macro F1-score** for balanced multiclass performance
- Tested key parameters:
  - **n_estimators:** 100–500
  - **max_depth:** None, 10, 20, 30
  - **min_samples_split:** 2, 5, 10
  - **min_samples_leaf:** 1, 2, 4
  - **max_features:** sqrt, log2
- Selected best model based on highest cross-validation score

Random Forest - Metrics



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nevus	0.62	0.61	0.62	75
squamous cell carcinoma	0.39	0.33	0.36	39
vascular lesion	0.73	0.68	0.70	28
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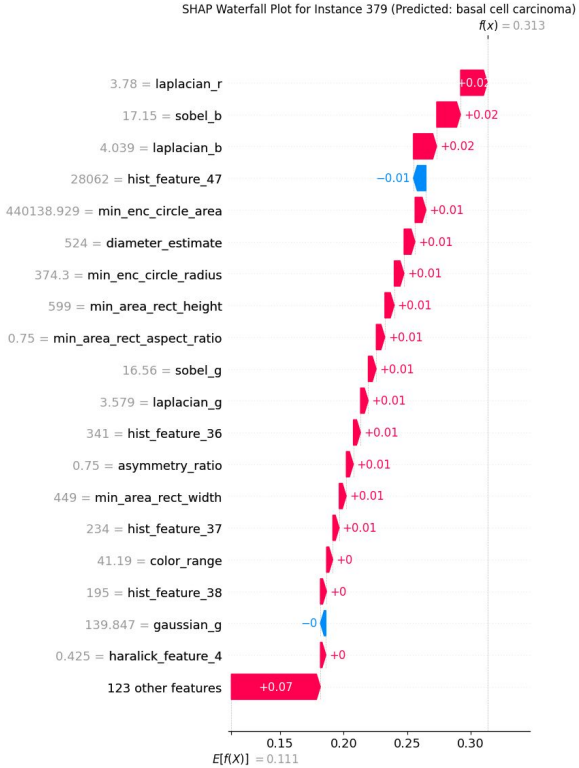
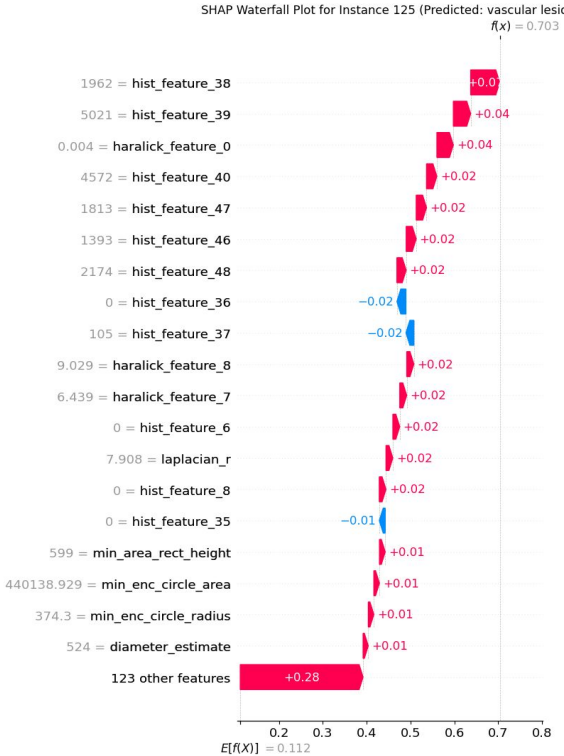
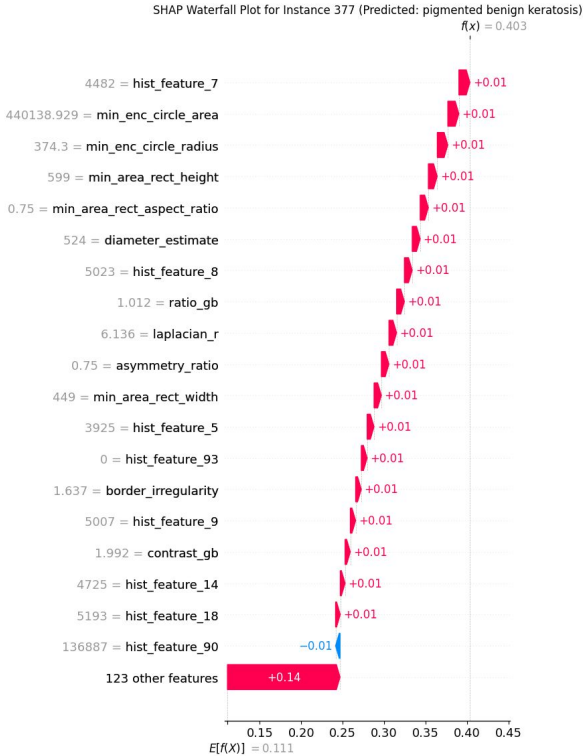
dermatofibroma	0.50	0.36	0.42	22
seborrheic keratosis	0.00	0.00	0.00	16
accuracy			0.60	472
macro avg	0.50	0.49	0.49	472
weighted avg	0.60	0.60	0.59	472

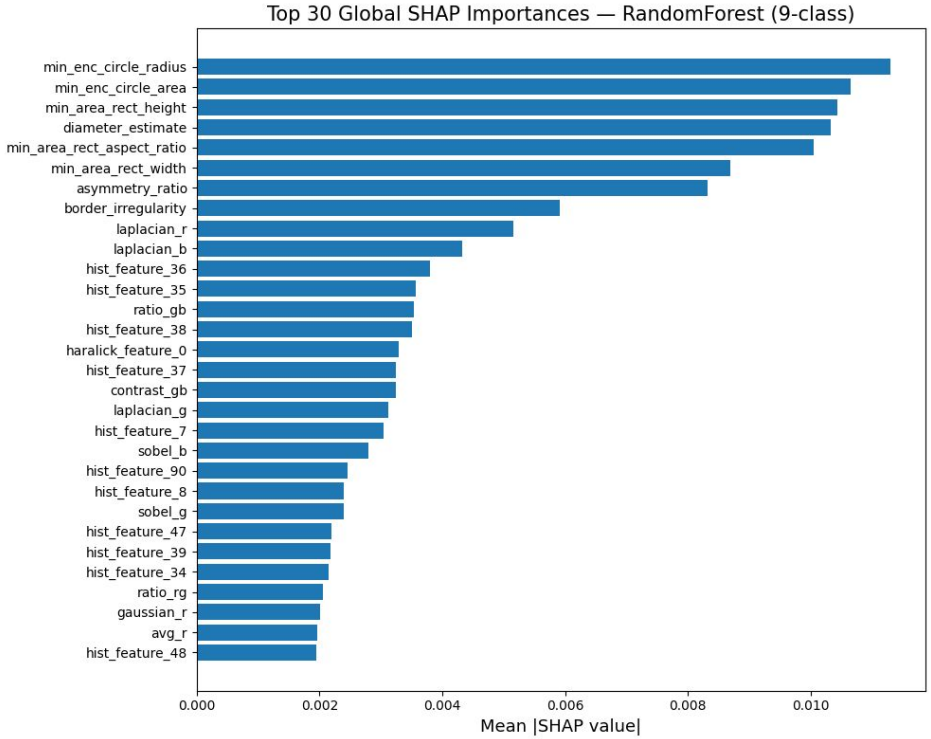
Random Forest - Local







Random Forest - Global



Insights

- Shape features dominated model decisions
- Size & asymmetry were strongest predictors
- Border irregularity improved lesion discrimination
- Texture/color had smaller impact

Limitations and Future Work

Limitations & Challenges

Insights

- Class imbalance made minority classes harder to predict accurately
- Overlapping class features caused confusion between similar categories
- Hyperparameter tuning required significant computation time
- Some misclassifications remained even after preprocessing and tuning
- Performance of the CNN model could be influenced by non-lesion features

Next steps

- XGBoost is the most promising model, but still not ready for deployment due to minority class misclassification
- Removing underrepresented classes with insufficient training samples to increase model performance
- Adding LBP texture features to the pipeline would provide finer texture detail for visually similar classes that current Haralick features fail to distinguish
- A hierarchical classifier could be used where a primary model identifies the broad lesion category and a secondary model resolves confusion between similar classes
- Alternative frameworks like LightGBM and CatBoost offer built-in class imbalance handling and may outperform XGBoost on minority classes without needing SMOTE.

THANKYOU

QUESTIONS ?